Example 9.57

The region enclosed by the circle $x^2 + y^2 = a^2$ is divided into two segments by the line $x = h$. Find the area of the smaller segment.

Solution

The smaller segment is sketched in Fig. 9.29. Here $0 < h < a$. By symmetry about the x-axis, the area of the smaller segment is given by

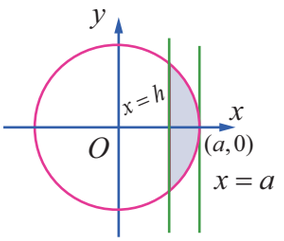


Fig. 9.29

$$A = 2\int_h^a \sqrt{a^2 - x^2}\, dx = 2\left[\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_h^a$$

$$= 2\left[0 + \frac{a^2}{2}\sin^{-1}(1)\right] - 2\left[\frac{h\sqrt{a^2 - h^2}}{2} + \frac{a^2}{2}\sin^{-1}\left(\frac{h}{a}\right)\right]$$

$$= a^2\left(\frac{\pi}{2}\right) - h\sqrt{a^2 - h^2} - a^2\sin^{-1}\left(\frac{h}{a}\right)$$

$$= a^2\left[\frac{\pi}{2} - \sin^{-1}\left(\frac{h}{a}\right)\right] - h\sqrt{a^2 - h^2}$$

$$= a^2\cos^{-1}\left(\frac{h}{a}\right) - h\sqrt{a^2 - h^2}\ .$$

Example 9.58

Find the area of the region in the first quadrant bounded by the parabola $y^2 = 4x$, the line $x + y = 3$ and y-axis.

Solution

First, we find the points of intersection of $x + y = 3$ and $y^2 - 4x$:

$x + y = 3 \Rightarrow y = 3 - x$.

$\therefore y^2 = 4x \Rightarrow (3 - x)^2 = 4x$

$\Rightarrow x^2 - 10x + 9 = 0$

$\Rightarrow x = 1, x = 9$.

$\therefore x = 1$ in $x + y = 3 \Rightarrow y = 2$, and $x = 9$ in $x + y = 3 \Rightarrow y = -6$.

$\therefore (1, 2)$ and $(9, -6)$ are the points of intersection.

The line $x + y = 3$ meets the y-axis at $(0, 3)$.

The required area is sketched in Fig. 9.30.

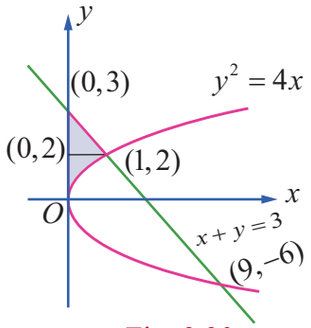


Fig. 9.30

Viewing in the direction of y-axis, on the right bounding curve is given by

$$x = \begin{cases} \dfrac{y^2}{4}, & 0 \le y \le 2 \\ 3 - y, & 2 \le y \le 3 \end{cases}$$

$$\therefore A = \int_0^2 x\, dy + \int_2^3 x\, dy = \int_0^2 \frac{y^2}{4}\, dy + \int_2^3 (3 - y)\, dy$$

$$= \left(\frac{y^3}{12}\right)_0^2 + \left(3y - \frac{y^2}{2}\right)_2^3 = \left(\frac{8}{12} - 0\right) + \left(9 - \frac{9}{2}\right) - \left(6 - \frac{4}{2}\right) = \frac{7}{6}.$$

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